Final exam · Unit 4 · Linear systems and phase portraits · 2.5 points · 50 min · Standard exam level

P18 — Damped mass-spring as a first-order system: the origin in the trace-determinant plane as the damping grows, and the critically damped motion through a generalised eigenvector

A body of mass $m = 1$ kg hangs from a spring of stiffness $k = 4$ N/m and moves inside a dashpot of damping coefficient $c \geq 0$ (N s/m). Its displacement $x(t)$ (m) from the equilibrium position obeys

$$m\ddot{x} + c\dot{x} + kx = 0.$$

(a) Write the equation as a first-order system $\dot{\mathbf{X}} = A\mathbf{X}$ for $\mathbf{X} = (x,\, v)^T$ with $v = \dot{x}$, for general $m, k > 0$ and $c \geq 0$. Locate A in the trace-determinant plane and classify the origin as a function of c: undamped, under-damped, critically damped and over-damped motion. With $m = 1$, $k = 4$ give the critical value c_{cr} and the eigenvalues for $c = 0$, $c = 2$ and $c = 5$, and describe each phase portrait in words.

(b) Take the critically damped case $c = c_{cr}$. Find the eigenvalue and eigenvector of A, a generalised eigenvector $\mathbf{w}$ with $(A - \lambda I)\mathbf{w} = \mathbf{v}$, the matrix $P = [\mathbf{v} \mid \mathbf{w}]$, and check explicitly that $P^{-1}AP$ is the Jordan form. Write the general solution. The body is released from $x(0) = 1$ m with velocity $v(0) = v_0$: find $x(t)$, the values of v_0 for which the body crosses the equilibrium position, and the crossing time for $v_0 = -6$ m/s. For $v_0 = 0$, find the largest speed reached.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $x(t)$ (m), independent variable $t \geq 0$ (s). Linear, homogeneous, constant coefficients, second order; $m, k > 0$, $c \geq 0$.
- Units: c/m and $\sqrt{k/m}$ are s^{-1}; k/m is s^{-2}. Eigenvalues are rates in s^{-1}.
- The coefficients are constant, so the existence and uniqueness theorem (Robinson Thm 11.1, Thm 25.2) gives a unique solution for every $(x(0), v(0))$, defined for all t.

Symbol	Value	Unit
m	1	kg
k	4	N/m
c	≥ 0, parameter	N s/m
$x(0)$	1	m
$v(0)$	v_0	m/s

Phase 2. Canonical formulation

With $x_1 = x$, $x_2 = v = \dot{x}$ (Robinson Sec. 25.1): $\dot{x}_1 = x_2$ and, dividing the equation by m, $\dot{x}_2 = -\frac{k}{m}x_1 - \frac{c}{m}x_2$:

$$\frac{d}{dt}\begin{pmatrix} x \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}\begin{pmatrix} x \\ v \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}.$$

Part (a) · 1 point

Classification of the origin as the damping varies

Step 1. Trace, determinant, characteristic polynomial

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{c}{m} - \lambda \end{vmatrix} = (-\lambda)\left(-\frac{c}{m} - \lambda\right) - 1 \cdot \left(-\frac{k}{m}\right) = \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m}.$$

It is the characteristic equation of the scalar ODE divided by m, as it must be. Hence $\operatorname{tr} A = -\frac{c}{m} \le 0$, $\det A = \frac{k}{m} > 0$ and

$$\lambda_{\pm} = \frac{-\frac{c}{m} \pm \sqrt{\frac{c^2}{m^2} - \frac{4k}{m}}}{2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

Step 2. The point A in the trace-determinant plane

$\det A = k/m$ is fixed and positive, so the origin is never a saddle. As c grows from 0, the point $(\operatorname{tr}, \det) = (-c/m,\ k/m)$ moves left along the horizontal line $\det = k/m$, starting on the det axis and crossing the parabola $\operatorname{tr}^2 = 4\det$ at $c/m = 2\sqrt{k/m}$, i.e. $c_{cr} = 2\sqrt{mk}$ (Robinson Ch. 31):

Damping	Position	Eigenvalues	Origin
$c = 0$ (undamped)	on the det axis	$\pm i\sqrt{k/m}$	centre (stable, not asymptotically)
$0 < c < c_{cr}$ (under-damped)	inside the parabola	$-\frac{c}{2m} \pm i\frac{\sqrt{4mk - c^2}}{2m}$	stable focus
$c = c_{cr}$ (critical)	on the parabola	$-\frac{c}{2m}$, double	stable degenerate node
$c > c_{cr}$ (over-damped)	outside the parabola	two negative reals	stable node

With $m = 1, k = 4$: $c_{cr} = 2\sqrt{1 \cdot 4} = 4$ N s/m.

Step 3. The three sample values

- $c = 0$: $\lambda^2 + 4 = 0$, $\lambda = \pm 2i$. Energy $\frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is conserved, so the orbits are the ellipses $4x^2 + v^2 = \text{const}$ (period π s). Test vector $\mathbf{X} = (1, 0)$: $A\mathbf{X} = (0, -4)$ points down, so the motion is **clockwise**.
- $c = 2$: $\lambda^2 + 2\lambda + 4 = 0$, $\lambda = \dfrac{-2 \pm \sqrt{4 - 16}}{2} = -1 \pm \sqrt{3}\, i$. Clockwise spirals into the origin (same test vector: $A(1, 0)^T = (0, -4)^T$), amplitude decaying as e^{-t}.
- $c = 5$: $\lambda^2 + 5\lambda + 4 = (\lambda + 1)(\lambda + 4) = 0$, $\lambda_1 = -1$, $\lambda_2 = -4$. Eigenvectors from row 1 of $(A - \lambda I)\mathbf{v} = \mathbf{0}$, $-\lambda v_1 + v_2 = 0$: $\mathbf{v}_1 = (1, -1)$, $\mathbf{v}_2 = (1, -4)$. Stable node; orbits reach the origin tangent to the slow direction $(1, -1)$ (smaller $|\lambda|$), and far away they are parallel to $(1, -4)$. The body returns to rest without oscillating, and it can cross $x = 0$ at most once.

KEY RESULTS

- $\lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$; discriminant $\dfrac{c^2 - 4mk}{m^2}$; $c_{cr} = 2\sqrt{mk} = 4$ N s/m.
- $c = 0$: centre, $\lambda = \pm 2i$. $0 < c < 4$: stable focus ($c = 2$: $\lambda = -1 \pm \sqrt{3}\, i$). $c = 4$: stable degenerate node, $\lambda = -2$ double. $c > 4$: stable node ($c = 5$: $\lambda = -1, -4$).
- Rotation in the (x, v) plane is clockwise in every oscillating case.

Part (b) · 1.5 points

Critically damped motion with a generalised eigenvector

Step 1. Eigenvalue and eigenvector

With $c = 4$: $\lambda^2 + 4\lambda + 4 = (\lambda + 2)^2 = 0$, so $\lambda = -2$ with algebraic multiplicity 2. Then

$$A + 2I = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix}.$$

Row 1: $2v_1 + v_2 = 0$; row 2: $-4v_1 - 2v_2 = -2(2v_1 + v_2) = 0$, the same equation. One free parameter: $\mathbf{v} = (1, -2)$. Since $A + 2I \neq 0$, there is no second independent eigenvector.

Step 2. Generalised eigenvector

Solve $(A+2I)\mathbf{w}=\mathbf{v}$: row 1 gives $2w_1+w_2=1$, row 2 gives $-4w_1-2w_2=-2$, which is row 1 times -2 (consistent). Choose $w_1=0$: $\mathbf{w}=(0,1)$. Check: $(A+2I)\mathbf{w}=(2\cdot 0+1,\ -4\cdot 0-2\cdot 1)=(1,-2)=\mathbf{v}$. Also $(A+2I)^2=\begin{pmatrix}4-4 & 2-2\\ -8+8 & -4+4\end{pmatrix}=0$, so $(A+2I)^2\mathbf{w}=\mathbf{0}$ as it must.

Step 3. Change of basis and Jordan form

$$P=[\mathbf{v}\mid\mathbf{w}]=\begin{pmatrix}1 & 0\\ -2 & 1\end{pmatrix},\qquad \det P=1\cdot 1-0\cdot(-2)=1,\qquad P^{-1}=\frac{1}{1}\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}.$$

Columns of AP: $A\mathbf{v}=-2\mathbf{v}=(-2,4)$ and $A\mathbf{w}=-2\mathbf{w}+\mathbf{v}=(0,-2)+(1,-2)=(1,-4)$; check directly: $A\mathbf{w}=(0\cdot 0+1\cdot 1,\ -4\cdot 0-4\cdot 1)=(1,-4)$. So $AP=\begin{pmatrix}-2 & 1\\ 4 & -4\end{pmatrix}$ and

$$\begin{aligned}(P^{-1}AP)_{11}&=1\cdot(-2)+0\cdot 4=-2, & (P^{-1}AP)_{12}&=1\cdot 1+0\cdot(-4)=1,\\ (P^{-1}AP)_{21}&=2\cdot(-2)+1\cdot 4=0, & (P^{-1}AP)_{22}&=2\cdot 1+1\cdot(-4)=-2,\end{aligned}\qquad P^{-1}AP=\begin{pmatrix}-2 & 1\\ 0 & -2\end{pmatrix}=J.$$

Step 4. Solution in the canonical coordinates

With $\mathbf{z}=P^{-1}\mathbf{X}$: $\dot z_2=-2z_2$ and $\dot z_1=-2z_1+z_2$. The first gives $z_2=c_2e^{-2t}$. The second is linear: multiply by the integrating factor e^{2t}, so by the product rule $\frac{d}{dt}(e^{2t}z_1)=e^{2t}(\dot z_1+2z_1)=e^{2t}z_2=c_2$. Integrate from 0 to t with Barrow's rule: $e^{2t}z_1(t)-z_1(0)=c_2\,[s]_0^t=c_2t$, so $z_1=(c_1+c_2t)e^{-2t}$ with $c_1=z_1(0)$.

Step 5. General solution

$\mathbf{X}=P\mathbf{z}=z_1\mathbf{v}+z_2\mathbf{w}$:

$$\mathbf{X}(t)=c_1e^{-2t}\begin{pmatrix}1\\ -2\end{pmatrix}+c_2e^{-2t}\left[t\begin{pmatrix}1\\ -2\end{pmatrix}+\begin{pmatrix}0\\ 1\end{pmatrix}\right],\qquad x(t)=(c_1+c_2t)e^{-2t},\quad v(t)=(-2c_1+c_2-2c_2t)e^{-2t}.$$

Check $v=\dot x$ by the product rule: $\dot x=c_2e^{-2t}+(c_1+c_2t)(-2)e^{-2t}=(c_2-2c_1-2c_2t)e^{-2t}$.

Step 6. Initial conditions

$x(0)=c_1=1, v(0)=-2c_1+c_2=v_0\Rightarrow c_2=v_0+2$:

$$x(t)=\big(1+(v_0+2)\,t\big)e^{-2t}.$$

Substitution check: with $x = (1+\beta t)e^{-2t}$, $\beta = v_0 + 2$, $\dot{x} = (\beta - 2 - 2\beta t)e^{-2t}$, $\ddot{x} = (-4\beta + 4 + 4\beta t)e^{-2t}$, and $\ddot{x} + 4\dot{x} + 4x = e^{-2t}\big[(4 - 4\beta + 4\beta t) + (4\beta - 8 - 8\beta t) + (4 + 4\beta t)\big] = 0$.

Step 7. Crossing the equilibrium

$e^{-2t} > 0$, so $x(t^*) = 0 \iff 1 + (v_0+2)t^* = 0 \iff t^* = -\dfrac{1}{v_0+2}$. This is a positive time only if $v_0 + 2 < 0$: the body crosses $x = 0$ (exactly once) iff $v_0 < -2$ m/s, i.e. it is thrown towards the equilibrium fast enough. For $v_0 = -6$: $t^* = -\frac{1}{-4} = 0.25$ s. For $v_0 \geq -2$ it approaches $x = 0$ from above without reaching it.

Step 8. Largest speed for $v_0 = 0$

$x = (1+2t)e^{-2t}$ and $v = \dot{x} = (2 - 2 - 4t)e^{-2t} = -4te^{-2t}$. Then $\dot{v} = (-4 + 8t)e^{-2t} = 0$ at $t = \frac{1}{2}$ s, a minimum of v (sign of $\dot{v}$ changes from $-$ to $+$). $|v|_{\max} = 4 \cdot \frac{1}{2}e^{-1} = \dfrac{2}{e} \approx 0.736$ m/s.

Step 9. Phase portrait

Degenerate stable node (Robinson Ch. 30): the only straight-line orbits lie on the eigenline $v = -2x$, with arrows pointing to the origin. Every other orbit is dominated by $c_2 t e^{-2t}\mathbf{v}$ both as $t \to +\infty$ (it enters the origin tangent to $v = -2x$) and as $t \to -\infty$ (it comes from far away parallel to the same line). Test vector $(1, 0)$: $A(1,0)^T = (0, -4)^T$, so orbits turn clockwise, and those that start above the eigenline with $v_0 < -2$ cross the v axis ($x = 0$) once before turning back towards the origin.

KEY RESULTS

- $A = \begin{pmatrix} 0 & 1 \\ -4 & -4 \end{pmatrix}$, $\lambda = -2$ (double), $\mathbf{v} = (1, -2)$, $\mathbf{w} = (0, 1)$.
- $P = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, $P^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, $P^{-1}AP = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}$.
- $x(t) = \big(1 + (v_0+2)t\big)e^{-2t}$; crosses $x = 0$ once iff $v_0 < -2$ m/s, at $t^* = -\dfrac{1}{v_0+2}$; $v_0 = -6$ m/s gives $t^* = 0.25$ s.
- $v_0 = 0$: $v(t) = -4te^{-2t}$, largest speed $2/e \approx 0.736$ m/s at $t = 0.5$ s.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to \infty$: $\mathbf{X} \to \mathbf{0}$ for all $c > 0$ (all eigenvalues have negative real part). Decay rate of the slowest mode, $\min|\mathrm{Re}\,\lambda|$: $c/(2m)$ below critical, 2 s^{-1} at $c = 4$, and $\frac{c - \sqrt{c^2 - 16}}{2} \to 0$ as $c \to \infty$ (very strong damping is slow). Critical damping gives the fastest return without oscillation. $c \to 0$: focus $\to$ centre.
- **Checks:** dimensions: $c_{cr} = 2\sqrt{mk}$ has $\sqrt{\text{kg} \cdot \text{N/m}} = \sqrt{\text{kg}^2\text{s}^{-2}} = $ kg/s = N s/m; $t^* = -1/(v_0+2)$: the 2 is 2 s^{-1} $\cdot$ 1 m, so $v_0 + 2$ is in m/s and $t^* = x(0)/|v_0+2|$ in s. Substitution of $x(t)$ in the ODE and of $\mathbf{X}$ in $\dot{\mathbf{X}} = A\mathbf{X}$ (script); $P^{-1}AP = J$.
- **Common mistakes in the exam:** (1) looking for a second eigenvector at $\lambda = -2$ and writing $c_1 e^{-2t}\mathbf{v} + c_2 e^{-2t}\mathbf{v}$, which is one solution twice; (2) writing the second solution as $te^{-2t}\mathbf{v}$ alone, without $\mathbf{w}$: it does not satisfy the system; (3) saying a critically damped body can never cross

equilibrium: it can, once, when $v_0 < -2$ m/s; (4) placing the over-damped case inside the parabola: the parabola $\mathrm{tr}^2 = 4\det$ separates foci (inside) from nodes (outside).